MAIS PAC

IPhO 02.12: Special Relativity · Worked Solutions

Special Relativity: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

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Solution 1 — The Lorentz factor. Return →

(a) $\gamma = 1/\sqrt{(1-0.36)} = 1/\sqrt{(0.64)} = 1.25$.

(b) Clocks run slow by 1.25 and lengths contract by the same factor: a 1 {m} rod measures 0.80 {m} along the direction of motion.

(c) $\gamma = 2$ requires $1-v^2/c^2 = 1/4$, so $v^2/c^2 = 3/4$ and $v = 0.866c$. **Check:** this is the value quoted in the text, and it is independent of the object — γ depends only on the speed.

Solution 2 — Muons from the upper atmosphere. Return →

(a) $\gamma = 1/\sqrt{(1-0.998^2)} = 1/\sqrt{(0.003996)} = 15.8$.

(b) The laboratory lifetime is $15.8\times2.2 = 34.8$ μ{s}, so the muon travels $0.998(3.0\times10\text{^}8)(34.8\times10\text{^}(-6)) = 1.04\times10\text{^}4$ {m} — just enough to reach the ground from 10 {km}.

(c) In the muon frame the atmosphere is moving at 0.998c and its thickness is contracted to 10,000/15.8 = 633 {m}. The muon lives 2.2 μ{s} and travels $0.998(3.0\times10\text{^}8)(2.2\times10\text{^}(-6)) = 659$ {m} in that frame — again just enough. **Check:** both frames agree that the arrival is marginal, which is why muon flux at sea level is so sensitive to the speed spectrum.

Solution 3 — A trip to a nearby star. Return →

(a) t = d/v = 4.3/0.80 = 5.4 years.

(b) γ = 1/√(1-0.64) = 1.667, so the proper time is t_0 = t/γ = 5.4/1.667 = 3.2 years.

(c) In the spacecraft frame the distance is L = L_0/γ = 4.3/1.667 = 2.6 light years, and t_0 = L/v = 2.6/0.80 = 3.2 years. **Check:** the two descriptions agree exactly, as they must: travelling 2.6 ly at 0.8c takes 3.2 years.

Solution 4 — Relativistic kinetic energy. Return →

(a) γ = 1/√(1-0.81) = 1/√(0.19) = 2.29, and mc² = (1.67×10^(-27))(9.0×10^(16)) = 1.50×10^(-10) {J}. So E = γ mc² = 3.44×10^(-10) {J}.

(b) K = (γ-1)mc² = 1.29(1.50×10^(-10)) = 1.94×10^(-10) {J} = 1.21×10³ {MeV}, i.e. about 1.2 {GeV}. The classical value (1)/(2)mv² = (1)/(2)(1.67×10^(-27))(0.81)(9.0×10^(16)) = 6.1×10^(-11) {J} = 380 {MeV} is less than a third of the truth.

(c) p = γ mv = 2.29(1.67×10^(-27))(0.90)(3.0×10^8) = 1.03×10^(-18) {kg,m,s^(-1)}. Then (pc)² = (1.03×10^(-18))²(9.0×10^(16)) = 9.6×10^(-20), (mc²)² = 2.3×10^(-20), and the sum is 1.19×10^(-19), whose square root is 3.4×10^(-10) {J} — the total energy. **Check:** for a proton, 1.2 {GeV} is roughly the rest energy 938 {MeV} times 1.3, so relativistic effects are comparable to the rest energy, as the numbers show.

Solution 5 — Mass into energy. Return →

(a) The reactor must produce 1.0/0.35 = 2.86 {GW} of thermal power, so the mass converted per second is

$$\frac{dm}{dt} = \frac{P}{c^2} = \frac{2.86\times10^{9}}{9.0\times10^{16}} = 3.2\times10^{-8}\ \{kg,s^{-1}\}. \tag{1}$$

(b) In a year (3.16×10^7 {s}) that is about 1.0 {kg}.

(c) The mass change is a kilogram out of thousands of tonnes of fuel, which is why it was missed for so long; but it releases the enormous energy that makes nuclear power practical, and the same relation explains why the Sun can shine for ten billion years. **Check:** 1 {kg} of mass gives 9×10^(16) {J}, and a 1 {GW} station produces 3×10^(16) {J} of electricity per year, so the two agree once efficiency is allowed for.

Solution 6 — Velocity addition. Return →

(a) u' = (u - v)/(1 - uv/c²) = (0.80 - (-0.60))/(1 + 0.48)c = (1.40)/(1.48)c = 0.946c. (Taking the probe as moving at +0.60c relative to the spacecraft, which moves at -0.80c relative to Earth in the spacecraft frame.)

(b) The Galilean answer is 1.40c, which is impossible. The denominator, which comes from the relativity of simultaneity, pulls the sum back below c.

(c) For u,v, write u = c(1-ε), v = c(1-δ) with ε,δ>0; then 1-uv/c² = ε+δ-εδ > 0 and u+v < 2c, so u' < c. **Check:** if either speed is exactly c, the result is c — the velocity of light is the fixed point of the addition law.

Solution 7 — The twin paradox. Return →

(a) The return trip is 18 light years at 0.90c: t = 18/0.90 = 20 years.

(b) γ = 1/√(1-0.81) = 2.294, so the travelling twin ages 20/2.294 = 8.7 years.

(c) The traveller accelerates at the turn-around: she is not in one inertial frame, while the stay-at-home twin is. The elapsed proper time along a world line is ∫√(1-v²/c²),dt, so the path with acceleration has the **larger** velocity averaged over the journey and accumulates **less** proper time — the travelling twin is genuinely younger. **Check:** an asymmetric difference of 11 years cannot be an artefact of a symmetric theory; the asymmetry is in the motion, not in the description.

Solution 8 — A relativistic particle in a magnetic field. Return →

(a) γ = 1/√(1-0.9025) = 3.20, and p = γ m_ev = 3.20(9.11×10^(-31))(0.95)(3.0×10^8) = 8.3×10^(-22) {kg,m,s^(-1)}.

(b) r = (p)/(qB) = (8.3×10^(-22))/((1.60×10^(-19))(1.0)) = 5.2×10^(-3) {m} = 5.2 {mm}.

(c) The classical radius m_ev/(qB) is 1.6 {mm} — a factor of γ = 3.2 too small. Accelerator designers must use the relativistic momentum, otherwise the magnets would be wrong by a factor of γ, which at LHC energies is thousands. **Check:** the period is still 2π m/(qB) times γ, so the cyclotron frequency falls: that is why synchrotrons ramp the frequency as the particles speed up.

Solution 9 — The lifetime of a particle beam. Return →

(a) $\gamma = 1/\sqrt{(1-0.9801)} = 7.09$.

(b) The laboratory lifetime is $7.09 \times 1.0 \times 10^{-8} = 7.1 \times 10^{-8}$ {s}, so the decay length is $0.99(3.0 \times 10^{8})(7.1 \times 10^{-8}) = 21$ {m}.

(c) In the particle frame the laboratory is moving at 0.99c, so the 21 {m} of laboratory is contracted to $21/7.09 = 3.0$ {m}; the particle travels $0.99(3.0 \times 10^{8})(1.0 \times 10^{-8}) = 3.0$ {m} in its own frame — the same length, as required. **Check:** both descriptions agree that the particle just survives 3 {m} of its own space, which is why beam pipes are designed around these numbers.

Solution 10 — Energy needed to reach a speed. Return →

(a) $\gamma = 2.294$, so $K = (\gamma-1)mc^2 = 1.294(1000)(9.0 \times 10^{16}) = 1.16 \times 10^{20}$ {J} — about 1.3 times the rest energy.

(b) $E = \gamma mc^2 = 2.294(9.0 \times 10^{19}) = 2.06 \times 10^{20}$ {J}.

(c) γ diverges as $v \to c$, so the energy needed diverges too: no finite amount of work can reach c. For scale, 1.16×10^{20} {J} is the entire world energy consumption for several thousand years, for one tonne of payload and no deceleration — which is the real obstacle to interstellar travel, not the speed limit itself. **Check:** at 0.99c the same spacecraft would need $\gamma = 7.09$, i.e. 5.5×10^{20} {J}, six times more for an extra 0.09c.